

# Financial Deserts in Latin America: A Subnational Atlas of Financial Exclusion

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## Abstract

Over 4,200 municipalities across Peru, Brazil, and Colombia—nearly half of all subnational units in the three countries—have zero bank offices. Using administrative data from financial regulators covering 8,582 districts and municipalities from 2002 to 2025, we construct the first subnational comparative atlas of physical financial access in Latin America and document three facts. First, financial deserts are pervasive: 73% of Peruvian districts, 46% of Brazilian municipalities, and 25% of Colombian municipalities lack any bank office as of 2024, affecting approximately 30 million people. Second, deserts are structural, not transitional: 96% of Peruvian districts that were deserts in 2012 remain deserts in 2024. Third, physical banking infrastructure is contracting across all three countries, with office counts declining 15–24% from peaks reached in 2014–2019. Rather than reaching underserved territories, the expansion of digital finance coincides with the closure of existing offices in already-served areas. These patterns suggest that prevailing financial inclusion strategies may accelerate the retreat of intermediation infrastructure from peripheral territories.

**Keywords:** financial inclusion, financial deserts, Latin America, territorial inequality, bank branches

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# 1 Introduction

Financial inclusion is widely recognized as a determinant of economic development and a dimension of inequality in its own right (Demirgüç-Kunt et al., 2018; Beck et al., 2007). Yet the dominant measures of financial access—the World Bank’s Global Findex and the IMF’s Financial Access Survey—operate at the national level, obscuring enormous subnational variation. A country may report 70% account ownership nationally while entire regions lack any financial infrastructure. This aggregation masks precisely the spatial inequalities that matter most for policy.

The concept of “deserts”—territories defined by the complete absence of a critical service—has been productive in public health, where “food deserts” identify areas lacking grocery stores and fresh food (Hegerty, 2020). We apply this logic to finance. A **financial desert** is a subnational unit with zero bank offices. The term captures a qualitative threshold: where physical financial infrastructure is entirely absent, residents cannot originate credit, access formal savings instruments, purchase insurance, or receive financial advisory—services that remain incompletely substitutable by digital channels or correspondent agents (Célérier and Matray, 2019; Burgess and Pande, 2005).

This paper introduces the Financial Deserts Atlas, the first comparative subnational database of physical financial access in Latin America. Covering Peru, Brazil, and Colombia from 2002 to 2025, the atlas documents three facts. First, financial deserts are pervasive: nearly half of all subnational units across the three countries lack any bank office. Second, deserts are persistent: once a territory becomes a desert, it almost never gains a bank office. Third, all three countries are experiencing a contraction in physical banking infrastructure—a reversal that coincides with the expansion of digital finance but concentrates closures in already-served areas rather than extending access to underserved ones.

These findings speak to a growing literature on the spatial determinants of financial access. Burgess and Pande (2005) showed that India’s social banking program, which mandated branch openings in unbanked locations, reduced rural poverty—establishing that physical bank presence has causal effects on economic outcomes. Bruhn and Love (2014) found similar impacts from Banco Azteca’s expansion into underserved Mexican municipalities. Célérier and Matray (2019) demonstrated that branch supply affects household financial inclusion even in high-income economies. Our contribution is to document that, across Latin America, the supply of physical banking infrastructure is not merely insufficient but actively declining.

The remainder of the paper proceeds as follows. Section 2 describes the data sources and methodology. Section 3 presents our results on the magnitude, persistence, population distribution, and temporal dynamics of financial deserts. Section 4 discusses the

implications of these patterns. Section 5 concludes.

## 2 Data and Methodology

We use administrative records from three national financial regulators to construct annual panels of bank office counts at the finest available subnational level.

For **Peru**, the Superintendencia de Banca, Seguros y AFP (SBS) provides the number of offices by institution type per district, annually from 2002 to 2025. Institution types include commercial banks (Banca Múltiple), municipal savings banks (Cajas Municipales), rural savings banks (Cajas Rurales), EDPYMES, and finance companies. Peru’s administrative structure comprises 1,891 districts, with a median population of 3,090 (INEI Census, 2017). The panel is balanced:  $1,891 \text{ districts} \times 24 \text{ years}$ .

For **Brazil**, the Banco Central do Brasil (BCB) publishes ESTBAN (Estatística Bancária por Município), a regulatory filing that records which bank agencies report deposits in each municipality each month. We use December snapshots from 2002 to 2024, accessed via [basedosdados.org](https://basedosdados.org). ESTBAN captures both openings and closures: an agency that stops reporting has effectively closed. Brazil has 5,570 municipalities; the panel is balanced ( $5,570 \times 23 \text{ years}$ ). Population data come from the IBGE Census (2022).

For **Colombia**, the Superintendencia Financiera (SFC) reports the number of offices per entity per municipality, quarterly from 2008 to 2025, available through the national open data portal ([datos.gov.co](https://datos.gov.co)). Colombia has 1,121 municipalities in our sample. Municipalities absent from SFC filings in a given year are coded as having zero offices. Population data come from the DANE Census (2018).

We define a **financial desert** as a subnational unit with zero bank offices in a given year. This definition captures access to full-service financial intermediation—credit origination, savings products, insurance, and advisory—which requires physical presence and is not fully substitutable by digital channels or agent correspondents (Allen et al., 2016; Kendall et al., 2010). We refer to this as a “Layer 1” definition, acknowledging that additional layers of access—correspondent agents, ATMs, mobile wallets—exist but serve distinct and more limited functions.

To complement the unit-based desert rate, we compute population-weighted desert rates using fixed census populations, which measure the share of the national population residing in financial deserts. This avoids conflating demographic change with changes in financial infrastructure and provides a more policy-relevant metric of exclusion.

Figure 1 illustrates the spatial distribution of financial deserts in Peru as of 2024, showing the concentration of deserts in Andean and Amazonian districts.

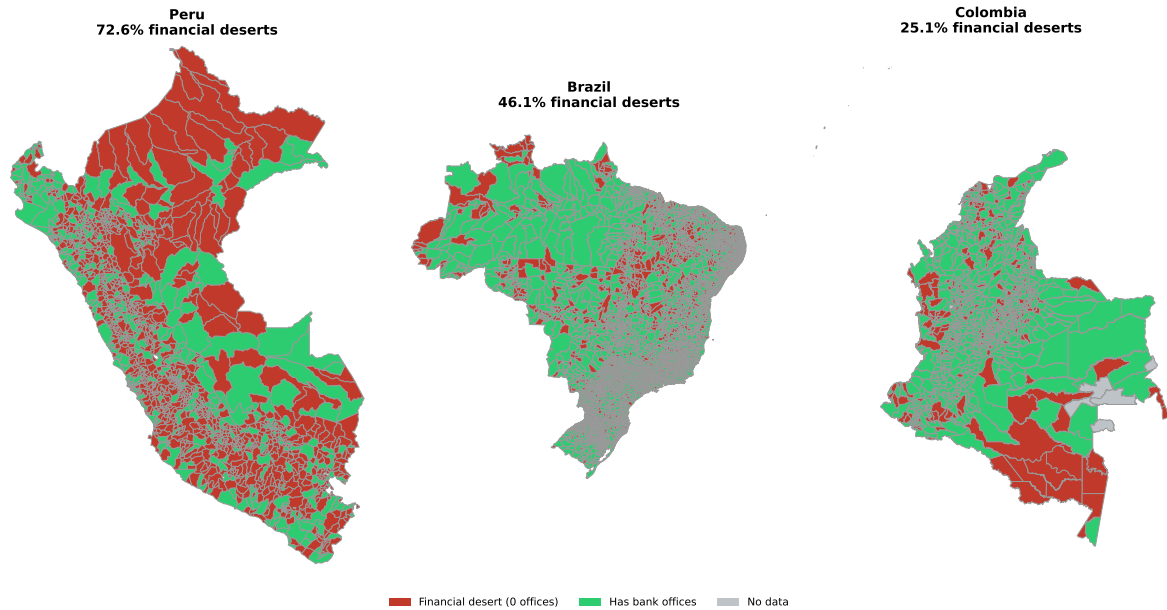


Figure 1: Financial deserts in Peru, Brazil, and Colombia, 2024. Red-shaded units have zero bank offices. Financial deserts concentrate in Andean highlands, Amazonian lowlands, and Brazil’s interior Northeast. An interactive version with year selection is available at <https://josemendozas.github.io/findeserts/>.

### 3 Results

Table 1 presents the cross-country snapshot for 2024. Nearly half of all subnational units across the three countries—4,220 districts and municipalities—have zero bank offices. Peru exhibits the highest desert rate at 72.6%, reflecting both the granularity of its administrative divisions and the extreme concentration of banking in coastal cities. Brazil’s 46.1% covers a far larger population base: over 2,500 municipalities without a single bank agency. Colombia, with a denser banking network relative to its administrative geography, still has one in four municipalities unserved.

Table 1: Financial Deserts in Latin America, 2024

| Country  | Units | With Office | Deserts | % Desert | Offices |
|----------|-------|-------------|---------|----------|---------|
| Peru     | 1,891 | 518         | 1,373   | 72.6%    | 3,986   |
| Brazil   | 5,570 | 3,004       | 2,566   | 46.1%    | 8,311   |
| Colombia | 1,121 | 840         | 281     | 25.1%    | 5,425   |
| Total    | 8,582 | 4,362       | 4,220   | 49.2%    | 17,722  |

These aggregate numbers, however, mask sharply different trajectories. Figure 2 shows the evolution of desert rates over time. Peru experienced a rapid reduction in financial deserts during the 2000s, driven by the expansion of Cajas Municipales and microfinance

institutions: the desert rate fell from 92% in 2002 to 75% by 2012, then stagnated. The districts reached during the expansion were predominantly provincial capitals with significant populations—the remaining deserts are smaller, more remote, and harder to serve profitably. Brazil’s trajectory is more alarming: a U-shaped curve. Brazilian financial deserts declined from 42% in 2002 to a minimum of 34% in 2013, then reversed, climbing back to 46% by 2024 as bank agencies closed faster than new ones opened. Colombia has remained stable at approximately 25% throughout 2008–2025.

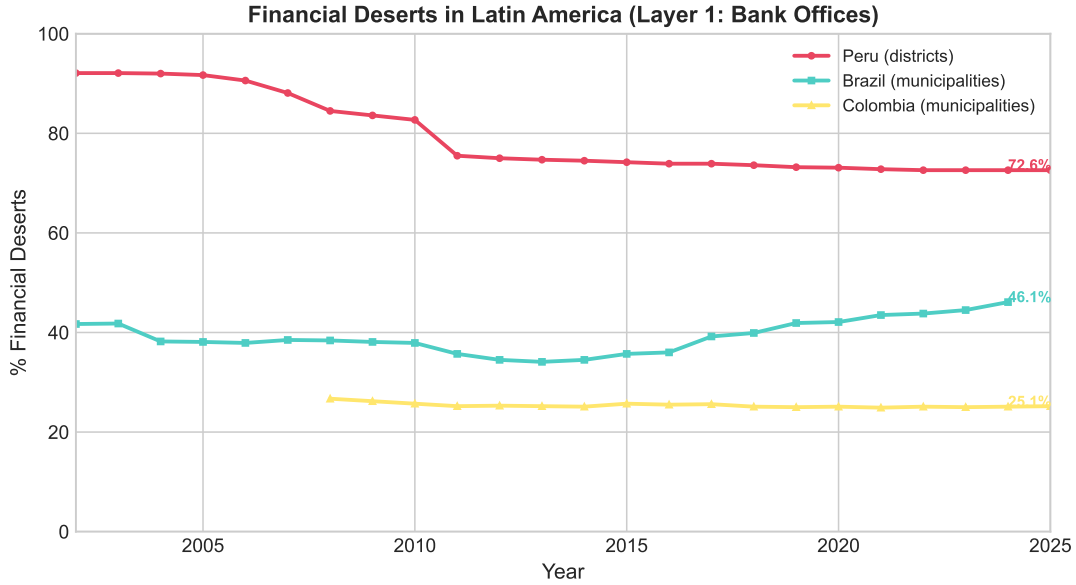


Figure 2: Percentage of subnational units classified as financial deserts, 2002–2025.

The persistence of financial deserts is striking. Table 2 reports transition matrices over the longest available window for each country. In Peru, 96% of districts that were deserts in 2012 remain deserts in 2024—only 53 districts gained a bank office over 12 years. In Brazil, persistence is lower (84.3%), but the direction of change compounds the concern: 607 municipalities *lost* their last bank agency between 2002 and 2024, substantially exceeding the 365 that gained one. Brazil is the only country in our sample experiencing net desertification. Colombia’s persistence rate of 88.6% is intermediate, with relatively few transitions in either direction.

Table 2: Persistence of Financial Desert Status

| Country  | Period    | Desert→Desert | Desert→Access | Access→Desert | Persistence |
|----------|-----------|---------------|---------------|---------------|-------------|
| Peru     | 2012–2024 | 1,366         | 53            | 7             | 96.3%       |
| Brazil   | 2002–2024 | 1,959         | 365           | 607           | 84.3%       |
| Colombia | 2008–2024 | 265           | 34            | 16            | 88.6%       |

This persistence raises a fundamental question: are financial deserts a consequence of in-

sufficient demand, or do they reflect coordination failures in which the absence of financial infrastructure depresses the economic activity that would justify entry? The literature on bank branching suggests the latter. Burgess and Pande (2005) found that mandated branch openings in unbanked Indian locations generated sufficient deposits to sustain operations, implying that latent demand existed but could not attract voluntary entry. The structural persistence we document—96% over 12 years in Peru—is consistent with an equilibrium trap in which low financial access and low economic complexity reinforce each other.

The unit-based desert rate, however, overstates the share of the population affected, because financial deserts tend to be small, sparsely populated territories. Figure 3 compares the unit-based and population-weighted desert rates for Peru and Brazil. While 72.6% of Peruvian districts are deserts, only 21% of the population lives in them. The gap reflects the success of Cajas Municipales and microfinance institutions in reaching district capitals with concentrated populations after 2005, even as the majority of districts—small, dispersed, and commercially unviable—remained unserved.

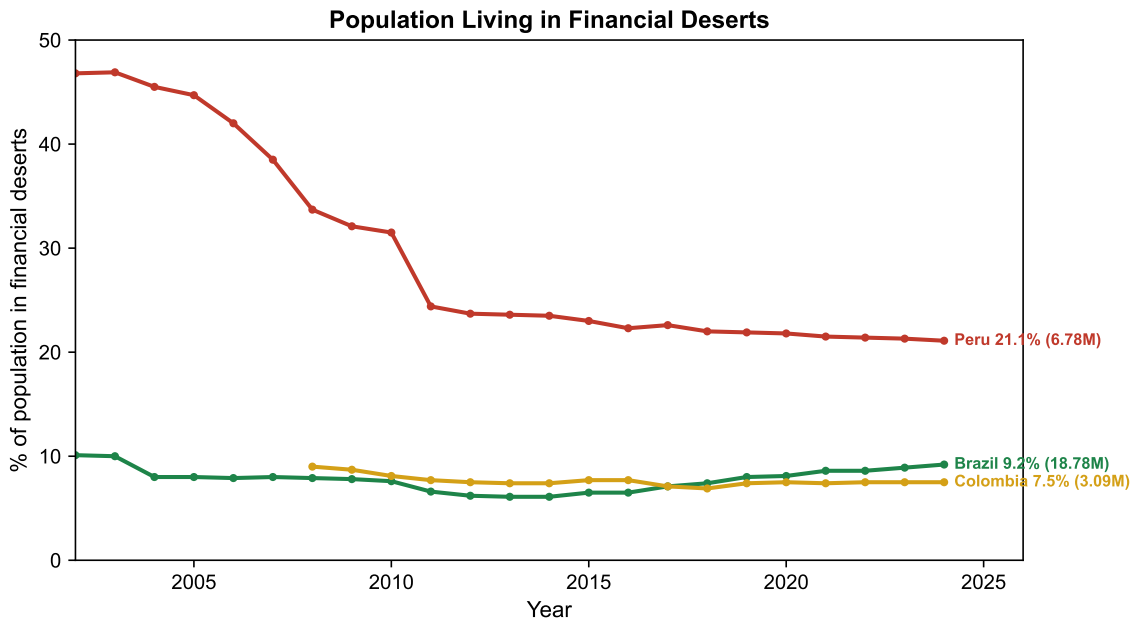


Figure 3: Share of national population living in financial deserts, Peru, Brazil, and Colombia. The population-weighted rate is substantially lower than the unit-based rate, but still affects approximately 30 million people across the three countries.

In Brazil, the population story reinforces the reversal narrative. The share of the population in deserts fell to 6.1% (12.4 million) in 2013 but has risen back to 9.2% (18.8 million) by 2024. Nearly 19 million Brazilians—nearly the population of Chile—now live in municipalities without a single bank agency. This is not a statistical artifact of administrative boundaries; it represents a large and growing population excluded from physical

financial intermediation.

Table 3: Population in Financial Deserts

| Country  | Census Pop. | Pop. in Deserts | % in Deserts | Desert Median Pop. |
|----------|-------------|-----------------|--------------|--------------------|
| Peru     | 32.2M       | 6.8M            | 21.1%        | 3,090              |
| Brazil   | 203.1M      | 18.8M           | 9.2%         | 5,737              |
| Colombia | 41.0M       | 3.1M            | 7.5%         | 4,850              |

The final dimension of our results concerns the contraction of physical banking infrastructure. All three countries have experienced declining bank office counts since peaks reached in 2014–2019. Figure 4 and Table 4 document the magnitude of this decline.

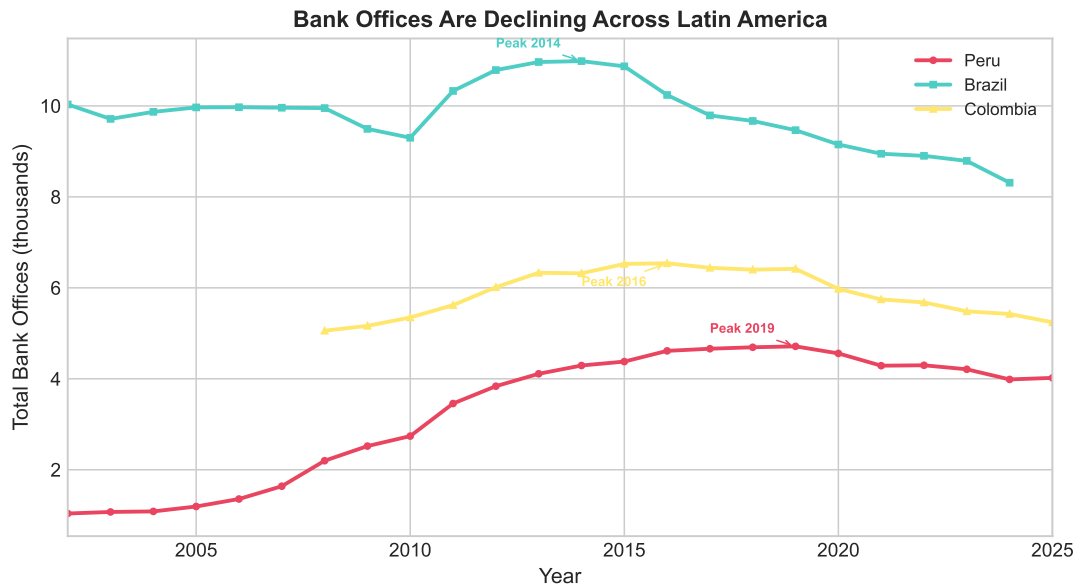


Figure 4: Total bank offices by country, 2002–2025. All three countries show declining office counts from their respective peaks.

Table 4: Bank Office Decline from Peak

| Country  | Peak Year | Peak Offices | 2024 Offices | Decline |
|----------|-----------|--------------|--------------|---------|
| Peru     | 2019      | 4,713        | 3,986        | −15.4%  |
| Brazil   | 2014      | 10,985       | 8,311        | −24.3%  |
| Colombia | 2016      | 6,540        | 5,425        | −17.0%  |

Brazil’s decline is the most severe: nearly one in four bank agencies has closed since 2014, coinciding with the rapid adoption of PIX (instant payments, launched 2020) and mobile banking. Peru’s decline accelerated after 2019, alongside the expansion of Yape and Plin. Colombia shows a 17.0% reduction since 2016. Crucially, these closures concentrate

in municipalities that *already had offices*, not in financial deserts. The net effect is a narrowing of the served territory. In Brazil, the 607 municipalities that lost their last bank agency between 2002 and 2024 outnumber the 365 deserts that gained one, producing a net increase in the number of financial deserts despite two decades of financial inclusion policy.

## 4 Discussion

The simultaneous expansion of digital finance and contraction of physical banking infrastructure across Latin America raises a question that existing literature has not adequately addressed: is the closure of bank offices an efficient market adjustment to changing technology, or does it represent a welfare-reducing retreat of financial intermediation from peripheral territories?

The efficiency interpretation holds that digital channels—mobile wallets, agent correspondents, instant payment systems—can substitute for physical bank offices, making brick-and-mortar infrastructure redundant. Under this view, the decline in office counts reflects rational cost optimization, and the affected populations are adequately served through alternative channels. There is evidence for this in urban and peri-urban contexts, where digital adoption is high and financial literacy sufficient for self-service ([Jack and Suri, 2014](#); [Sahay et al., 2020](#)).

The market-failure interpretation, however, is more consistent with the patterns we document. If digital channels were adequate substitutes, we would expect desert rates to decline as offices close—that is, digital access would compensate for physical withdrawal. Instead, we observe stable or increasing desert rates alongside office contraction. This suggests that the populations in financial deserts are not effectively reached by digital alternatives. The persistence of deserts—96% over 12 years in Peru—indicates that private entry does not occur even as potential markets grow. This is consistent with coordination failures of the type identified by [Burgess and Pande \(2005\)](#): without an initial anchor institution, the fixed costs of entry exceed the expected returns from thin local markets, even though aggregate welfare gains from access may be substantial ([Bruhn and Love, 2014](#)).

The distinction between transactional inclusion and intermediation inclusion is critical here ([Cull et al., 2014](#); [Allen et al., 2016](#)). Peru has over 166,000 correspondent agent points (agentes) reaching 99% of districts, and mobile wallets like Yape have achieved near-universal urban penetration. By transactional metrics—the ability to send and receive payments—Peru’s financial inclusion has improved dramatically. But correspondent agents cannot originate loans, evaluate credit risk, or provide financial advisory. The closure of bank offices eliminates the local capacity for intermediation, potentially locking



peripheral economies into a low-complexity equilibrium in which financial services are limited to cash-in/cash-out transactions (Dupas et al., 2018).

The regional implications are significant. Financial deserts in all three countries are not randomly distributed: they concentrate in territories that are also disadvantaged along other dimensions—indigenous populations, limited connectivity, dependence on subsistence agriculture (Beck et al., 2007). The retreat of physical banking from these territories risks compounding existing spatial inequalities, creating a two-tier financial system in which metropolitan populations access the full range of intermediation services while peripheral populations access only basic payment functionality (Kendall et al., 2010).

Our analysis has several limitations. The administrative level differs across countries: Peru’s 1,891 districts are more granular than Brazil’s 5,570 municipalities or Colombia’s 1,121, which mechanically inflates Peru’s desert rate in cross-country comparisons. The definition of financial deserts as “zero offices” is binary and does not capture variation in the quality or range of services available in municipalities with one or few offices. Finally, we do not model the determinants of desert status—the interplay of population density, distance to urban centers, regulatory policy, and institutional strategy—which would require a different empirical framework.

## 5 Conclusion

The evidence presented here suggests that prevailing financial inclusion strategies in Latin America—centered on digital payments and agent correspondents—may be insufficient to address the structural exclusion of peripheral territories from financial intermediation. If bank offices continue to close without effective substitutes for credit origination and financial advisory, the result will be an expansion of transactional access alongside a contraction of intermediation access. Future work should examine whether regulatory interventions—branching mandates, differentiated licensing for underserved areas, public banking models—can reverse the desertification process, and whether digital lending platforms can eventually substitute for the intermediation functions that physical offices currently provide. The atlas, available at <https://josemendozas.github.io/findeserts/>, provides the empirical foundation for this research agenda.

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